GIT TROUBLESHOOTING AND REAL-WORLD ISSUES

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1. Git push rejected — non-fast-forward

Issue: Another developer pushed changes to GitHub before you pushed your code.

1. git pull --rebase origin main
2. git push origin main

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2. Resolve merge conflicts properly

Add the complete resolution workflow:

1. git status
   # Edit the conflicted file
2. git add <filename>
3. git commit

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3. Accidentally committed a secret or password

Issue: An AWS access key, database password, or .env file was pushed to GitHub.

- Revoke or rotate the exposed credential immediately.
- Add secrets and .env files to .gitignore.
- Understand that deleting a file in a later commit does not remove the secret from Git history.
- Learn the basics of history cleanup and secret scanning.

Important: Never rely on .gitignore to remove files that Git already tracks.

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4. GitHub authentication and permission errors

Issue: Permission denied (publickey) or Authentication failed.

1. git remote -v
2. ssh -T git@github.com
3. git config --global user.name
4. git config --global user.email

Explain SSH keys, HTTPS authentication with a personal access token where applicable, repository permissions, and the difference between commit identity and authentication.

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6. Recover deleted commits or lost work

Issue: A student resets the wrong commit or accidentally deletes a branch.

1. git reflog
2. git log --oneline --all
3. git branch recovery <commit-id>

Explain how git reflog can help locate recent commit references and how to create a recovery branch before attempting further changes.

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7. Undo the wrong commit safely

- git restore — discard selected working-tree changes.
- git reset — move the current branch or unstage changes, depending on the options.
- git revert — create a new commit reversing an earlier commit.
- git commit --amend — update the most recent commit; use caution if it has already been shared.

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8. Pull request cannot be merged

Issue: A PR has conflicts, failed checks, or insufficient approvals.

First check how to inspect the PR, update the feature branch, resolve conflicts, rerun checks, and request the required review.

Explain protected branches and why direct pushes to production branches should be restricted.

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9. Local branch and remote branch are out of sync

First to inspect differences before integrating changes:

1. git fetch origin
2. git status

3. git log --oneline --graph --decorate --all
4. git diff main origin/main

Explain ahead/behind commits, divergent histories, and why git fetch is useful before deciding whether to merge or rebase.

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10. .gitignore is not ignoring a file

Issue: A file such as .env continues to appear in Git status.

1. git check-ignore -v .env
2. git ls-files .env
3. git rm --cached .env

Explain the difference between an ignored file and a file already tracked by Git. Removing a file from tracking does not erase its previous commits.

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11. Git merge and rebase failures

what happens when a rebase stops because of conflicts and how to continue or cancel it.

1. git status
   Resolve the conflicted files
2. git add <filename>
3. git rebase --continue
   # To cancel instead:
4. git rebase --abort

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12. src refspec main does not match any

how to troubleshoot a wrong branch name or a repository that has no commit yet.

1. git status
2. git branch
3. git add .
4. git commit -m "Initial commit"
5. git branch -M main
6. git push -u origin main

Explain that renaming the branch alone does not create the initial commit; students must commit their files first.

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13. Detached HEAD state

Explain why Git may show a detached HEAD after checking out a commit or tag, and how to preserve new work.

1. git status
2. git switch -c recovery-branch

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14. git bisect

What is git bisect to narrow down the commit that introduced a regression.

1. git bisect start
2. git bisect bad
3. git bisect good <known-good-commit>

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# Test the checked-out version
git bisect good

# Or, if the bug exists:
git bisect bad

# Repeat until Git identifies the faulty commit
git bisect reset
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